Interpretable by Construction, Identifiable by Construction: Privacy Leakage of Shapelet-Based Anti-Money-Laundering Models

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ABSTRACT

Investigators and supervisors ask that anti-money-laundering (AML) models be explainable, and shapelet classifiers are a natural answer: they justify a decision by pointing to a short, human-readable pattern in an account's activity. We show that this explanation is frequently *a customer's own transaction history*. On SAML-D (9.5M transactions, 28 typologies) we train shapelet detectors on activity-matched weekly account series and hand the fitted model to an adversary holding ∼30k candidate accounts. For the classic shapelet transform, 15% of released shapelets uniquely re-identify their source customer from shape alone, and 66% of those sources are flagged launderers, so the model also discloses who is under suspicion. For the state-of-the-art Random Dilated Shapelet Transform (RDST), which stores per-shapelet normalisation statistics, 86% of shapelets single out exactly one customer (≈827 customers) and reveal their exact weekly amounts and counts. The leak survives a realistic adversary whose amounts are off by ±5% (227 customers re-identified at 96% precision per 300 shapelets) and degrades only when the adversary misses a large share of transactions. It replicates on AMLworld (63% of RDST shapelets single out one customer). Standard membership-inference audits miss it (AUC ≈ 0.50–0.52); conventional models leak *whether* a customer was in training, if anything, while shapelet models leak *who* they are and *what* they did. Noise and k-anonymous averaging remove exact fingerprints at no accuracy cost but leave residual linkage, whereas *learned* shapelets match RDST accuracy (PR-AUC 0.957 vs. 0.948) without copying any customer's data, and show no memorisation even when overtrained on small data. Interpretable financial models need a privacy review of their explanations, not only of their predictions.

1 INTRODUCTION

Transaction monitoring for AML is high-stakes, regulated, and increasingly machine-learned; deployed systems pair gradient-boosted trees with post-hoc explanations so that investigators can act on alerts [7]. Shapelets [1, 2] are an attractive alternative because the explanation *is* the model: a short subsequence whose presence drives the decision. Laundering typologies such as structuring, fan-in/fan-out and rapid pass-through are literally shapes in time.

The catch is that classic shapelets are subsequences copied from training series, and state-of-the-art variants such as RDST [4] additionally store the mean and standard deviation used to normalise each shapelet. Federated shapelet work notes that shapelets expose raw data [8], and privacy-aware matrix profiles defend shared time-series features against pattern inference [10]; explanation-privacy research shows that feature attributions leak membership [11]. To our knowledge, no prior work measures what a *released shapelet model* reveals about specific customers.

Contributions. (i) A threat model and attack suite—re-identification, amount reconstruction, membership and suspicion inference—for released shapelet models. (ii) An empirical study on two AML benchmarks showing large but *concentrated* leakage that aggregate membership inference misses, including disclosure of flagged customers (a tipping-off hazard), and mapping when it survives an imperfect adversary. (iii) A privacy–utility comparison of mitigations showing that learned shapelets attain near state-of-the-art shapelet accuracy without copying customer data.

2 THREAT MODEL

A bank trains a shapelet AML detector and shares the fitted model with a vendor, auditor, regulator sandbox, or consortium partner, or loses it to theft. The model contains its shapelets (and, for RDST, their normalisation statistics). The adversary also holds candidate account series, e.g. a partner institution's data or a leaked extract, containing training members and non-members. We study both an idealised adversary (exact copy, full account list) and a realistic one whose amounts are perturbed, who sees only part of each account's transactions, and whose list holds only some accounts. Goals: (a) *re-identify* the customer behind a shapelet; (b) *reconstruct* that customer's amounts and counts; (c) infer *membership* and, from the shapelet's class, *suspicion status*.

3 EXPERIMENTAL SETUP

Data. SAML-D [6]: 9.5M synthetic transactions over 46 weeks, 0.10% suspicious, 17 suspicious and 11 normal typologies. Each account becomes a 6-channel weekly series (log outgoing/incoming amount, outgoing/incoming count, cross-border amount, cash amount). Accounts involved in any laundering transaction are positive (7,902). Normal accounts are sampled 3:1 *matched on number of active weeks*; unmatched normals are mostly single-week accounts that make detection trivial. The 29,946 accounts are split 50/50 by account into members (train) and non-members (test), stratified by label; results average 3 seeds. **AMLworld** HI-Small [9] (IBM's NeurIPS 2023 benchmark) spans only ∼10 days, so we use 6-hour bins (40 steps); 23,409 activity-matched accounts, 5,862 positive.

Detectors. CatBoost on 40 aggregate features plus the flattened series (the industry baseline); MiniRocket [5] with logistic regression; the binary shapelet transform (5,000 random non-constant z-normalised candidates, quality = class separation of the min-distance profile, top-200 with self-similarity removal) with ExtraTrees; RDST (1,000 shapelets) with ExtraTrees; and learned shapelets [3] (144 shapelets, soft-min distance, random initialisation).

Attacks. *Re-ID*: min (dilated) sliding distance from each released shapelet to every candidate account; we report the share of shapelets whose source is the *only* account within ε (unique match) and top-1 accuracy. *Reconstruction*: for RDST, raw window = values × std + mean. *MIA*: score = −min distance to any shapelet; for baselines, the standard confidence attack. *Realistic adversary*: we

Table 1: Detection on held-out accounts and leakage of the released model (mean ± std over 3 seeds). Re-ID: share of released shapelets that match *only* their source customer among 29,946 candidate accounts.

Model	PR-AUC	R@1%FPR	MIA AUC	Re-ID (%)
CatBoost (aggregates)	0.983 ± 0.001	0.842 ± 0.031	0.504 ± 0.002	n/a
MiniRocket	0.955 ± 0.002	0.637 ± 0.044	0.508 ± 0.003	n/a
Shapelet transform	0.878 ± 0.007	0.493 ± 0.033	0.500 ± 0.002	15.0 ± 2.2
RDST	0.948 ± 0.006	0.631 ± 0.021	0.520 ± 0.001	85.6 ± 1.6
RDST, norm.-only, no stats	0.925 ± 0.005	0.515 ± 0.015	0.512 ± 0.000	74.9 ± 1.9
Learned shapelets	0.957 ± 0.004	0.667 ± 0.017	0.500 ± 0.000	0.0

Table 2: Mitigations (shapelet transform; mean over seeds). Unique: shapelet matches only its source; Top-1: source is the nearest account.

Release	PR-AUC	Unique (%)	Top-1 (%)
As trained	0.878 ± 0.007	15.0 ± 2.2	19.3 ± 1.6
Noise σ=0.1	0.879 ± 0.007	0.0 ± 0.0	10.8 ± 2.5
Noise σ=0.25	0.877 ± 0.007	0.0 ± 0.0	3.2 ± 0.5
Noise σ=0.5	0.876 ± 0.005	0.0 ± 0.0	0.2 ± 0.2
Noise σ=1.0	0.870 ± 0.004	0.0 ± 0.0	0.0 ± 0.0
k-anon. mean, k=5	0.877 ± 0.007	0.3 ± 0.2	13.5 ± 0.4
k-anon. mean, k=20	0.876 ± 0.005	0.2 ± 0.2	12.5 ± 2.1
k-anon. mean, k=50	0.876 ± 0.006	0.0 ± 0.0	10.7 ± 1.3

Table 3: Re-identification by a realistic adversary (SAML-D, 300 sampled RDST shapelets, mean over seeds). The adversary's copy of the data has perturbed amounts and/or misses transactions, and holds all or half of the accounts; it claims an identity only if the nearest account is at least 2× closer than the runner-up. Cust.: customers correctly re-identified; Prec.: share of claims that are correct.

	All accounts		Half of accounts	
Adversary's view	Cust.	Prec.	Cust.	Prec.
Exact copy	251	94%	125	86%
Amounts ±5%	227	96%	115	86%
80% of txns, ±5%	71	78%	36	62%
50% of txns, ±5%	10	27%	5	17%

rebuild the adversary's series from the raw transactions with log-normal amount noise (σ=0.05) and each transaction kept with probability 0.8 or 0.5, subsample the candidate list, and claim an identity only if the nearest account is ≥ 2× closer than the runner-up.

4 RESULTS

Leakage is severe but concentrated. Table 1: on SAML-D only the shapelet models show leakage; CatBoost and MiniRocket, which store no training subsequences, give no membership signal. Because just a few hundred of ~15k members contribute shapelets, aggregate MIA AUC stays near 0.5, yet those contributors are uniquely identifiable: an audit based on standard MIA would declare these models safe.

Table 4: Second dataset: AMLworld HI-Small (23,409 activity-matched accounts, 6-hour bins; mean over 1 seeds). MIA: standard membership-inference AUC. Re-ID: share of shapelets matching only their source; Cust.: customers singled out per model.

Model	PR-AUC	MIA AUC	Re-ID (%)	Cust.
CatBoost	0.624	0.623	n/a	n/a
MiniRocket	0.479	0.587	n/a	n/a
Shapelet transform	0.389	0.497	5.0	–
RDST	0.571	0.505	62.9	606

Shape alone identifies. Z-normalised shapelet-transform shapelets carry no amounts, yet 15% match only their source among ~30k accounts (top-1: 19%).

State of the art leaks more. RDST is the most accurate shapelet model here but stores normalisation statistics, so 86% of its shapelets reconstruct an exact, unique slice of one customer's history: about 827 customers per model, each with a median of 13 exact non-zero weekly amount/count cells (median peak weekly amount ≈ £25k). Withholding the statistics and forcing every shapelet to be normalised barely helps: 75% of shapelets still match a single customer (PR-AUC drops to 0.925), because a multi-channel dilated shape is itself a near-unique fingerprint—and once an account is re-identified, the adversary's own data supplies its amounts.

A realistic adversary still succeeds (Table 3). Perturbing every amount by ±5% barely matters: the adversary re-identifies 227 customers (62 flagged) at 96% precision per 300 RDST shapelets; with stored statistics withheld, 193 at 94%. With half the account list, it finds about half as many, still mostly correctly. The limit is coverage of each account's activity: seeing 80% of transactions leaves 71 correct identifications at 78% precision, while at 50% precision falls to 27%. The threat is therefore real for parties that see an account's full activity (vendors and auditors given bank data, insiders, breaches) and much weaker for outsiders with a partial view.

It replicates on a second benchmark (Table 4). On AMLworld, with 6-hour bins and a much harder task, RDST is the best time-series detector (PR-AUC 0.57; CatBoost 0.62) and 63% of its shapelets single out one customer (≈606 per model, 144 flagged); shape-only shapelets leak less (5%). Here CatBoost overfits and does leak membership (MIA AUC 0.62), which sharpens the contrast: a conventional model leaks *whether* a customer was used, a shapelet model leaks *who* they are and *what* they did, and the standard audit sees only the former.

The model discloses suspicion. 66% of shapelet-transform sources and 217 RDST-identified customers are positives. Releasing the model therefore reveals which customers were flagged, in tension with suspicious-activity-report confidentiality and anti-tipping-off rules.

Mitigations (Table 2). Gaussian noise on shapelet values removes exact fingerprints at no accuracy cost, but nearest-neighbour linkage persists until $\sigma \geq 0.5$. Replacing each shapelet by the mean of its k nearest training windows from distinct accounts (k-anonymous release) stops unique matching but keeps the source's neighbourhood linkable. Learned shapelets dominate: PR-AUC 0.957 (shapelet transform: 0.878; CatBoost: 0.983), with no learned shapelet within

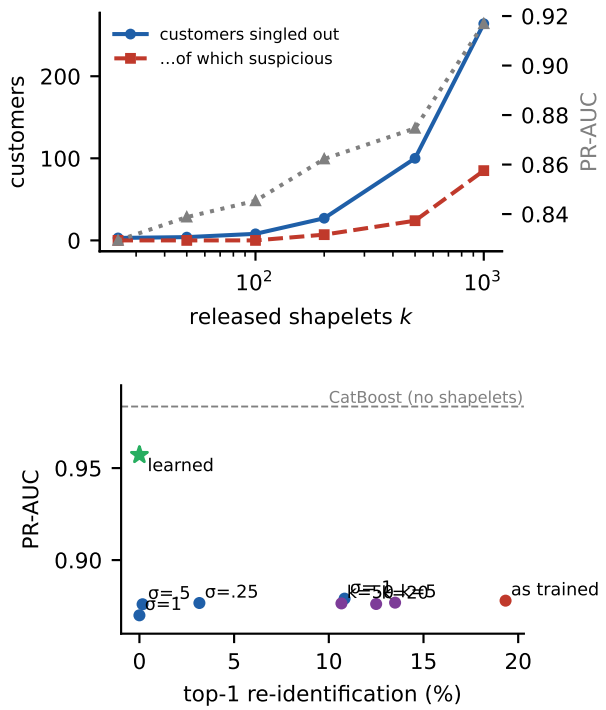


Figure 1: Top: customers uniquely exposed (and how many are flagged launderers) grow with the number of released shapelets k, while accuracy saturates (shapelet transform, seed 0). Bottom: privacy–utility of releases (top-1 re-identification vs. PR-AUC; seed mean).

10^{-2} of any real account (median nearest distance 0.38). Pushing them to memorise—up to 400 epochs on as few as 300 training accounts—does not change this: in every setting the share of learned shapelets whose nearest account is a training member equals the members' share of the pool (chance), and the adversary makes 0 confident claims.

5 DISCUSSION AND LIMITATIONS

SAML-D is synthetic and activity-matched sampling makes detection optimistic; however, the leakage mechanism (copied subsequences, stored statistics) does not depend on the data. Our shapelet transform is a faithful simplification (AUC-based quality instead of information gain) because a standard implementation degenerated on our sparse series. Our realistic-adversary results bound the threat: it needs a near-complete view of each targeted account. A stronger, per-example membership audit (e.g. shadow-model likelihood-ratio attacks) may detect some of this leakage; our claim concerns the standard aggregate audit. Noise and learned shapelets carry no formal guarantee; differentially private shapelet selection and DP-SGD training are natural next steps.

Recommendations. Treat shapelets as personal data; do not rely on withholding normalisation statistics; prefer learned or noised shapelets; and audit interpretable financial models with *per-explanation* re-identification, not only aggregate membership inference.

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